Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/29/2005
PI Name:	Akins, Robert E. Ph.D.		
Project Title:	Cell:Cell Mediated Assembly of Cardiac Tissue in Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rakins@nemours.org	**Fax:**	FY 302-651-6897
PI Organization Type:	NON-PROFIT	**Phone:**	302-651-6811
Organization Name:	A.I. duPont Hospital for Children		
PI Address 1:	Department of Biomedical Research		
PI Address 2:	P.O. Box 269		
PI Web Page:	http://www.udel.edu/		
City:	Wilmington	**State:**	DE
Zip Code:	19899	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	03/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Kirby, Mary	**Contact Phone:**	
Contact Email:	mary.f.kirby@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
	This project addresses aspects of cardiac tissue engineering. The capacity of isolated rat and human heart cells to reconstitute cardiac tissue in culture without extensive cues from three-dimensional scaffolds or matrices is being investigated. The approach takes full advantage of the simulated-microgravity environment provided by NASA-bioreactors to allow isolated cells to re-establish tissue structure in vitro. The proposed research also investigates the role of gravity and the bioreactor environment in the structural and molecular/biochemical phenotype of cardiac cell assemblages. A combination of morphological, cell biological, molecular, and biochemical methods will be used to characterize cardiac cell cultures and to compare bioreactor-derived cultures to native tissue and to cultures grown using standard tissue culture methods. In the long-term, it is hoped that this study will lead to the ability to engineer biosynthetic cardiac patches for the treatment of a wide variety of cardiac abnormalities. Further, it is hoped that this research will lead to an increased understanding of the influence of gravity on cells and cell assemblages in culture. The work investigates the ability of isolated heart cells to re-establish structural and molecular/biochemical phenotypes of		

Task Description:	heart tissue when cultured in microgravity environments. In particular, the study will: 1)Evaluate the process by which cardiac-specific structures are formed by isolated rat and human cardiac cells. 2)Identity the extra-cellular matrix (ECM) material that is differentially produced in bioreactor-based cultures. 3) Assess the role of gravity in determining the distribution of subcellular organelles in cultured heart cells. 4) Evaluate the effect of bioreactor-based culture on signal transduction pathways. 5)Determine the changes in gene expression caused by culture in microgravity using differential display RT-PCR. For Spaceflight: The long-term presence of humans in space will require that critical medical and surgical procedures be conducted at remote sites utilizing limited resources. Of particular concern will be the ability to maintain adequate circulatory function, which is required to sustain life and to allow crew-members to remain productive during flight and thereafter. Among the issues of relevance to NASA's spaceflight programs, two are critical. First, physical and pharmacologic countermeasures that allow crew members to remain productive during shifting hemodynamic-loading conditions associated with changes in gravity are needed; strategies that require limited additional "upweight" would be especially useful. Our studies focus on pathways that impact the organization and function of cardiac muscle in order to develop pharmacologic interventions that will improve circulatory function under changing conditions or in disease. We have identified a number of potentialyy importnat pathways including a heme-metabolic pathway that may be amenable to pharmacologic manipulations using simple and naturally-occurring compounds. Stimulation of this pathway improves cardiac muscle cell organization and function. Second, there is a need to develop strategies that reverse or repair profound heart disease that may arise in crew members. Heart disease, which is a leading cause of morbidity and mortality in humans, may occur as a direct result of the spaceflight environment, as a secondary result of accommodation to long-term mission conditions, or as a consequence of unanticipated disease, injury, or infection. Our studies are directed toward the development of implantable constructs produced on site using defined media with primary isolated cells and simple bioreactors. Thus, although at an early stage, our research may be highly relevant to the development of strategies supporting the long-term presence of humans in space. On Earth: Congenital and acquired heart diseases are significant problems. The treatments that are currently available mainly address the sequelae of primary defects and often fall short of full structural and functional recovery. In addition, there is a severe shortage of donor tissue for reconstructive cardiac surgery and transplantation. Advances in treatment are hampered by a lack of suitable material and a lack of characterization of the molecular pathways that establish and sustain tissue structure. New treatments for heart diseases, new sources of implant materials, and accessible models for the study of cardiac morphogenetic pathways are all needed, and tissue-engineering has emerged as a potential means to address these areas. Cardiac tissue engineering is an emerging field involving the construction of tissue equivalents in vitro. Using an approach in which self-assembly is encouraged, work by the applicant shows that isolated neonatal rat heart cells (NRHCs) re-establish characteristics of three-dimensional cardiac tissue when grown in specialized bioreactors. Through the use of NASA biotechnologies, our knowledge of the mechanisms and pathways involved with in vitro tissuegenesis is being extended so that an accessible source of tissue-equivalent material can be developed. The development of tissue-equivalents for use as research models or for use as implant devices could profoundly alter the course of cardiac-related research and dramatically improve the outlook for patients with a variety of heart diseases.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Results during the 3.5 years of study funded by NAG9-1339 establish a framework for the mechanistic understanding of de novo cardiac tissuegenesis and of the role of heme metabolic pathways in modulating cardiac muscle phenotype. Crucial steps have been taken toward developing strategies to manipulate cardiac cells, to improve cardiac performance, and to repair damaged cardiac tissue. Further studies are needed to understand the behavior of cardiac cells and cardiac stem cells in microgravity environments, to probe the mechanisms by which heme oxygenase-mediated pathways improve cardiac organization and function while protecting against oxidative injury, and to assess the application of these emerging therapeutic strategies in models of heart disease. This work represents the initial phase in the development of advanced counter-measures to enhance cardiac performance and to repair damaged cardiac tissue using approaches that are suitable for long-term NASA missions.
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